GREEN-MCS: A Data-Driven Dynamic Dimensioning Framework for Energy-Efficient SDN Failover

Kevin Saavedra, Dario Guaman, Dayana Delgado
Department of Informatics, Modeling, Electronics and Systems (DIMES)
University of Calabria (UNICAL), Rende, Italy

February 28, 2026

Abstract

Software-Defined Networking (SDN) enables flexible failover orchestration, yet carrier-grade infrastructures suffer from excessive energy consumption due to the static over-provisioning of high-performance routing hardware. Balancing strict sub-50 ms Service Level Agreements (SLAs) with green networking initiatives remains a critical mathematical challenge. This paper introduces GREEN-MCS, a data-driven, multi-objective dynamic dimensioning framework that optimizes the Pareto frontier between total network power consumption and recovery latency. By decoupling logical failover roles from physical hardware capacities, the algorithm dynamically maps network nodes to either low-power (Zodiac FX) or high-performance (NEC PF5240) profiles based on historical peak traffic and M/M/1 queuing constraints. To validate its resilience, the framework is subjected to rigorous algorithmic simulation across continental (Abilene), regional (Nobel-Germany), and national (Germany50) topologies under varying degrees of stochastic traffic variance (σ). Results demonstrate that topological redundancy fundamentally governs green efficiency. In dense national meshes, GREEN-MCS achieves a massive 73.1% maximum energy reduction while maintaining ultra-low sub-4 ms recovery delays. Furthermore, the framework features an autonomous "Safety Valve" mechanism; under extreme stochastic burst stress ($\sigma = 700$), it guarantees up to 20.4% continuous power savings in regional meshes by preserving redundant low-power zones, while safely defaulting to legacy hardware in rigid backbones to prevent catastrophic queuing collapse. Ultimately, GREEN-MCS proves that carrier-grade reliability and aggressive energy conservation are not mutually exclusive when orchestrated through physics-aware, topology-sensitive models.

Contents

1 Introduction **3**
1.1 Contributions . . . 3

2 Objectives **5**
2.1 General Objective . . . 5
2.2 Specific Objectives . . . 5

3 Related Work **6**

4 GREEN-MCS SDN Design **7**
4.1 Software and Hardware Specifications . . . 7
4.2 Base Topologies . . . 8
4.3 Software Architecture & Operational Parameters . . . 9

5 GREEN-MCS SDN Implementation **11**
5.1 Phase I: Data Acquisition & Modeling . . . 11
5.1.1 Geo-Spatial Graph and SPF Routing . . . 11
5.1.2 Stochastic Normalization and Peak Extraction . . . 11
5.1.3 Operational Stress Parameters . . . 12
5.1.4 Hardware Profiling and Energy Models . . . 12
5.2 Phase II: GREEN-MCS Algorithmic Engine . . . 12
5.2.1 Data-Driven Hardware Profiling . . . 12
5.2.2 Combinatorial Set Discovery and Heuristic Expansion . . . 13
5.2.3 Pure Recovery Delay and M/M/1 Physics . . . 13
5.3 Phase III: Multi-Objective Optimization and Output . . . 14
5.3.1 Multi-Objective Scoring and Optimal Placement . . . 14
5.3.2 Algorithm Workflow . . . 14
5.3.3 Iterative Failure Injection and Topology Stressing . . . 15
5.3.4 Stochastic Parameter Sweeping . . . 15
5.3.5 Data Aggregation and Output Profiling . . . 16

6 Results and Discussion **17**
6.1 Abilene Topology Analysis . . . 17
6.2 Nobel-Germany Topology Analysis . . . 19
6.3 Germany50 Topology Analysis . . . 22
6.4 Cross-Topology Benchmarking: Scalability and Resilience Summary . . . 24
6.5 Discussion . . . 25

7 Conclusion and Future Work **27**
7.1 Conclusions . . . 27
7.2 Limitations and Future Work . . . 27

1 Introduction

The exponential growth of global data traffic has positioned the Information and Communication Technology (ICT) sector as a major consumer of electricity, necessitating a paradigm shift towards energy-aware network infrastructures. In traditional carrier-grade networks, the standard design practice relies on the homogeneous deployment of high-capacity, chassis-based switches to guarantee peak performance [1]. However, this "over-provisioning" strategy leads to substantial energy wastage, as high-performance devices often operate significantly below their packet processing capacity during off-peak periods or in low-demand topological segments [2].

While Software-Defined Networking (SDN) has enabled dynamic traffic engineering through logical centralization, the physical power consumption of the network remains bound by the underlying hardware specifications. Recent studies indicate that a purely homogeneous deployment creates a dichotomy: utilizing exclusively high-performance switches (e.g., core routers) is inherently inefficient for edge nodes, whereas deploying exclusively low-power devices risks violating throughput requirements and Service Level Agreements (SLA) due to saturation [3].

To resolve this inefficiency, this project introduces **GREEN-MCS**, a data-driven, multi-objective framework for heterogeneous SDN failover orchestration. Unlike transitional hybrid IP/SDN models [4], our approach assumes a fully SDN-enabled environment composed of diverse physical profiles: legacy high-performance switches (NEC PF5240) and energy-efficient forwarding devices (Zodiac FX). The core innovation of Green-MCS lies in its traffic-aware resource allocation and the decoupling of the logical recovery plane from physical hardware limits. Rather than relying on static over-provisioning, the algorithm predictively evaluates stochastic Packet Per Second (PPS) traffic variance (σ) against strict M/M/1 queuing bounds, dynamically assigning power-hungry hardware exclusively to structural bottlenecks. Ultimately, this methodology mathematically maps the Pareto frontier between aggregate energy conservation and sub-50 ms carrier-grade recovery latency, demonstrating that green networking and absolute operational resilience can seamlessly coexist.

1.1 Contributions

While previous works, such as Vuppalapati et al. [4], laid the foundation for progressive network recovery using homogeneous hardware, this research addresses the critical limitations of static, topology-agnostic assumptions in carrier-grade environments. The specific contributions of the **GREEN-MCS** framework are four-fold:

- **Data-Driven Heterogeneous Dimensioning:** We transition from the standard homogeneous paradigm to a predictive, heterogeneous SDN model. By decoupling the logical orchestration plane from physical hardware limits, the framework dynamically provisions a hybrid infrastructure—combining high-performance legacy switches (NEC PF5240) with low-power edge devices (Zodiac FX)—based on stochastic traffic variance (σ) rather than static over-provisioning.

- **Physics-Aware SLA Enforcement:** We advance beyond abstract graph-theory connectivity by integrating stochastic M/M/1 queuing physics directly into the optimization logic. This guarantees that selected failover anchor nodes ("Heroes") are not only topologically optimal but physically capable of absorbing extreme packet-per-second (PPS) micro-bursts without inducing buffer saturation or violating the strict sub-50 ms recovery SLA.

- **Multi-Objective Pareto Optimization (α):** We introduce a tunable routing policy parameter (α) that maps the Pareto frontier between aggregate power consumption and network latency. This novel control mechanism allows operators to dynamically scale the active control-plane overhead (by expanding or compressing the optimal set K of active

Heroes) to prioritize absolute green efficiency or maximum performance based on the network's physical state.

- **Cross-Topological Scalability Analysis:** We provide an exhaustive empirical evaluation across continental (Abilene), regional (Nobel-Germany), and national (Germany50) topologies. This benchmarking uncovers fundamental behavioral laws, such as *Topological Gravity* and the *Green Penalty Paradox*, demonstrating mathematically that a network's geographical path redundancy natively dictates its ecological potential and operational resilience.

2 Objectives

2.1 General Objective

To design and validate **Green-MCS**, a data-driven orchestration framework for heterogeneous SDN architectures that dynamically balances high-performance and low-power hardware. By integrating M/M/1 queuing constraints with a tunable multi-objective optimization mechanism, the system minimizes global energy consumption while mathematically guaranteeing proactive resilience against single-link failures and strict adherence to sub-50 ms Service Level Agreements (SLAs).

2.2 Specific Objectives

- **Network Resilience via Optimal "Hero" Selection::** To ensure network availability against single-link failures by implementing a proactive-reactive recovery mechanism. The system identifies the minimal set of "Hero" nodes capable of protecting the entire logical topology, dynamically adapting to topological changes.
- **Physical Viability Modeling (M/M/1 Awareness):** To surpass abstract graph-theory connectivity by integrating physical hardware constraints into the optimization engine. The framework models queuing delays under extreme traffic variance (σ) using the relationship between arrival rates (λ) and CPU-bound service capacities (μ), guaranteeing that recovery paths never exceed the 50 ms SLA limit.
- **Energy Efficiency (Green Networking):** To reduce total infrastructure power consumption through a heterogeneous hardware deployment strategies. The goal is to maximize the utilization of low-power devices (Zodiac FX $\approx$ 15W) for recovery tasks, limiting the activation of high-power hardware (NEC PF5240 $\approx$ 118W) strictly to nodes with high traffic demands.
- **Multi-Objective Pareto Tuning (α):** To provide a deterministic control mechanism that allows network administrators to dynamically shift the operational priority. Utilizing the tunable routing parameter (α), the framework seamlessly transitions across the Pareto frontier, trading active control-plane overhead (Hero set size K) for absolute energy savings or maximum recovery speed.
- **Scientific Validation and Benchmarking:** To quantify the performance of the proposed heterogeneous solution against a standard baseline (All-NEC deployment). The objective is to demonstrate that a "Green" resilient network can be achieved without compromising latency-critical *Service Level Agreements* (SLA).

3 Related Work

The foundational logic for resilient node selection is derived from Vuppalapati and Venkatesh [4], who proposed the *Minimum Candidate Selection* (MCS) algorithm. Their work applies a greedy set-cover approach to identify the smallest subset of nodes required to recover from Single Link Failures (SLF). In our Green-MCS framework, we adapt this mathematical principle to identify "Hero nodes"—critical points in the topology that must handle aggregated recovery traffic.

The architectural trade-offs in network design were extensively analyzed by Galán-Jiménez [2]. His research on genetic algorithms demonstrated that replacing a strategic subset of nodes with specialized hardware could yield energy savings comparable to fully optimized networks. We leverage this concept of "Strategic Deployment," shifting the focus to a Heterogeneous Hardware Assignment problem where the optimization variable is the class of the SDN switch.

Crucial to the physical realism of our approach is the precise characterization of hardware packet processing limits. For high-capacity nodes, we rely on the technical specifications of the **NEC PF5200 series** [5], which documents a maximum throughput of 131 Million Packets Per Second (Mpps) and non-blocking switching fabrics. This data serves as the upper bound (μ_{NEC}) in our M/M/1 queue models for core aggregation points.

Conversely, for energy-efficient edge nodes, we utilize the architectural analysis of the **Zodiac FX** presented by Schiff et al. in the *Loko* project [6]. As detailed in their hardware breakdown (Figure 1), the device relies on an Atmel SAM4E8E Cortex-M4 microcontroller running at **120 MHz**. Unlike the ASIC-based NEC switches, the Zodiac's packet processing is CPU-bound. By correlating the CPU clock cycles with the Ethernet frame overheads analyzed in their work, we derive a realistic, constrained service rate (μ_{Zodiac}) for the low-power nodes.

Finally, the quantification of energy efficiency is grounded in the *RESDN* metric proposed by Assefa and Özkasap [1]. Their work provides validated empirical power consumption profiles for these specific devices ($\approx$ 118W for NEC, $\approx$ 15W for Zodiac). Green-MCS combines these power profiles with the aforementioned throughput constraints to execute a traffic-aware orchestration that prevents saturation while minimizing energy.

4 GREEN-MCS SDN Design

This section details the algorithmic setup and methodological framework used to evaluate the GREEN-MCS system. First, Section 4.1 defines the software routing simulation and the hardware power models adopted for realistic energy quantification. Next, Section 4.2 introduces the carrier-grade SNDLib topologies used to test the framework's scalability. Finally, Section 4.3 outlines the core software architecture and the operational stress parameters—specifically stochastic packet variance (σ) and burst multipliers (B_m)—applied to validate QoS compliance under highly dynamic, non-deterministic traffic conditions.

4.1 Software and Hardware Specifications

To ensure the reproducibility of the experiments and the accuracy of the energy consumption models, we define the specifications for both the execution environment (Host) and the simulated network entities (Emulated Profiles).

Host Hardware Requirements
The experimental framework is designed to be lightweight yet robust. It runs on a standard CPU-based architecture, eliminating the need for specialized ASIC hardware. The project is fully hosted and version-controlled on GitHub[1], making it accessible for community replication. For optimal performance, especially when emulating larger topologies, we recommend the following host configuration:

- **Environment:** A Linux-based Virtual Machine (VM) or dedicated machine running Ubuntu (20.04 LTS recommended).
- **Resources:** A quad-core CPU and a minimum of 8 GB RAM are required to support the concurrent execution of the SDN controller and the Mininet topology without memory bottlenecks.

Emulated Network Hardware Profiles
A critical aspect of this study is the use of realistic power profiles. While the experiment runs virtually, the energy metrics are calculated based on empirical data from physical hardware specifications.

- **NEC PF5240 (Standard SDN):** A commercial OpenFlow switch modeled with a base consumption of 118.33 W and a dynamic load of 0.52 W per active port [1].
- **Zodiac FX (Green SDN):** An ultra-low power OpenFlow switch designed for efficiency, consuming only 15 W base and 0.15 W per port [1].

Software Stack
The project relies on a cohesive open-source ecosystem to ensure modularity and reproducibility. The implementation is orchestrated within an Ubuntu Linux environment, utilizing Python as the primary programming language. The specific components include:

- **Development Environment:** The logic was developed using Visual Studio Code, executing on an Ubuntu 20.04 LTS platform to ensure compatibility with network namespaces and system-level dependencies.
- **Mininet:** The industry-standard network emulator employed to instantiate the virtual data plane and reproduce the topological switching fabric.

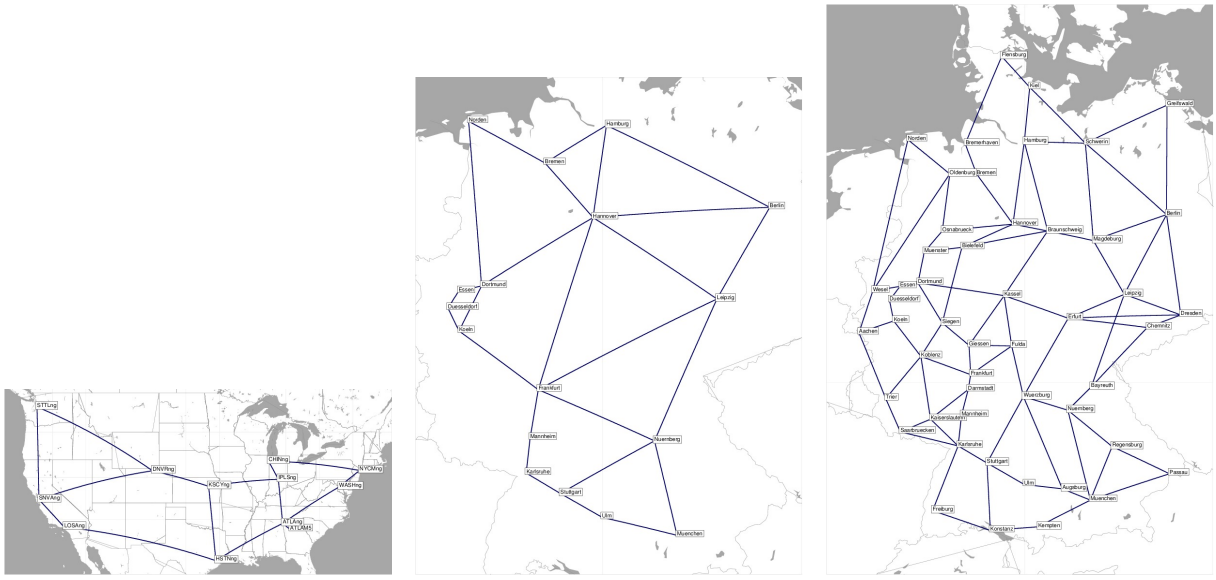
[1] https://github.com/Saavkd7/SDN-IP/tree/graficas

- **Ryu Controller:** A Python-based SDN control framework that manages flow rules, topology discovery, and rule installation via the OpenFlow 1.3 protocol.
- **Algorithmic Core:** The Python codebase leverages *NetworkX* for graph theoretic operations and *NumPy* for numerical analysis of the α-based trade-offs.
- **SNDLib:** The topologies (Abilene, Nobel-EU, Germany50) are parsed directly from the Survivable Network Design Library to ensure topological accuracy.

4.2 Base Topologies

To validate the GREEN-MCS algorithm, we utilize standard network topologies from the Survivable Network Design Library (SNDLib)[2]. A critical constraint of the proposed optimization logic is the fundamental requirement for physical path redundancy. Consequently, the algorithm is explicitly designed for mesh and ring topologies. It is structurally incompatible with tree or star networks due to the inherent presence of leaf nodes (i.e., terminal nodes with a degree of one). In such hierarchical structures, the failure of a terminal link completely isolates the leaf node from the rest of the network, making it physically and mathematically impossible for the SDN controller to establish an alternative recovery path.

Although the library provides numerous topologies (e.g., *cos266*, *nobel-eu*), many lack the comprehensive traffic matrices required to calculate the packet-per-second switching load necessary for our heterogeneous hardware assignment. Therefore, our experimental evaluation focuses on three specific topologies that offer both the required structural complexity and reliable traffic data:



(a) Abilene (Modified) (b) Nobel-Germany (c) Germany50

Figure 1: Selected SNDLib mesh topologies for heterogeneous SDN evaluation.

1. **Abilene (Modified):** Originally an 11-node backbone, this topology requires specific preprocessing for our algorithmic validation. To ensure graph biconnectivity suitable for the Green-MCS logic, the topology is adapted by eliminating a specific node that creates a cut-vertex, resulting in a robust testbed for failure recovery scenarios.
2. **Nobel-Germany:** A regional network topology that provides a dense set of 26 interconnections for 17 nodes; ideal for testing the algorithm's ability to balance NEC and Zodiac

[2] https://sndlib.put.poznan.pl/home.action

placements in a highly meshed environment.

3. **Germany50:** A large-scale national network consisting of 50 nodes. This topology serves as the primary stress test for scalability, evaluating how the algorithm assigns heterogeneous resources under high-density constraints.

4.3 Software Architecture & Operational Parameters

The core of the GREEN-MCS framework is modularly structured into three sequential phases. To accurately model the highly dynamic nature of real-world carrier networks, the architecture processes not only the raw topological data but also a set of critical operational variables that simulate physical network stress:

- **Stochastic Packet Variance (σ):** Real network traffic does not consist of fixed-size payloads. We introduce σ to model the standard deviation around an average IMIX packet size, bounded by standard Ethernet limits. This allows the framework to mathematically assess how an influx of small packets exponentially increases the Packet Per Second (PPS) load on the physical forwarding engines.

- **Burst Multiplier (B_m):** Standard traffic matrices provide data averaged over fixed intervals (e.g., 5 minutes), which inherently masks instantaneous micro-bursts. To prevent the algorithm from optimistically assigning low-power switches based on flat averages, B_m is applied to scale the baseline traffic, forcing a worst-case scenario stress test on the network queues.

- **Multi-Objective Weight (α):** The core tuning parameter ($\alpha \in [0, 1]$) that dictates the framework's optimization priority, seamlessly shifting the algorithmic focus between absolute energy conservation and strict recovery latency minimization.

This pipeline is specifically designed to transition from these stochastic inputs to a fully optimized, SLA-compliant controller placement scheme. Figure 2 details the internal processing flow across its three main operational phases.

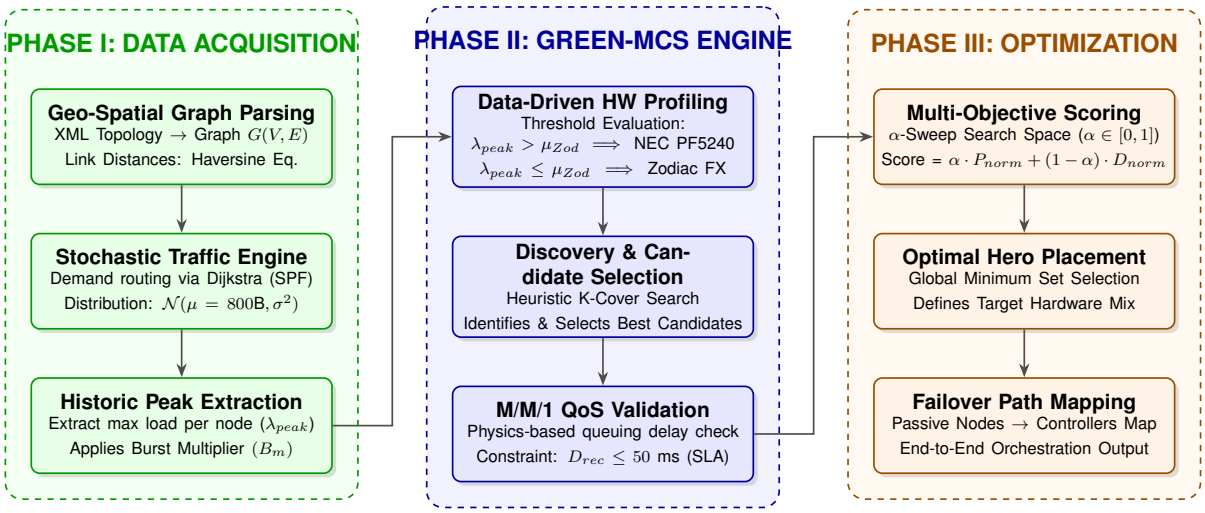


Figure 2: GREEN-MCS Software Architecture Pipeline. The workflow integrates Data Acquisition and Predictive Modeling (Phase I),GREEN-MCS Algorithmic Engine (Phase II), and Multi-Objective Optimization (Phase III).

Phase I - Data Acquisition and Modeling
The workflow begins by translating the abstract network topology into a mathematically rigorous environment. The framework parses the XML files, applying Haversine formulas to establish accurate geographical link weights (distances) between nodes. Subsequently, it utilizes a Stochastic Traffic Engine to simulate demand routing via Dijkstra's Shortest Path First (SPF) algorithm. By applying extreme byte variance (σ^2) and burst multipliers (B_m), the engine extracts the historic peak load (λ_{peak}) for every individual switch. Calculating this maximum expected transit load *a priori* is the fundamental enabler of the framework; it establishes a deterministic upper bound of traffic ingestion for each node before any failure simulation occurs.

Hardware Processing Constraint (The Zodiac FX Profile): To contextualize the traffic loads extracted in Phase I, the framework utilizes strict hardware definitions. The low-power profile (Zodiac FX) is designed for edge efficiency with a baseline power of 15.0 W. Crucially, its processing capacity (μ_{Zod}) is strictly constrained to 100 kpps. This threshold is mathematically derived from its 120 MHz embedded CPU. In software-based switching, the theoretical maximum packet processing rate (μ_{max}) is defined by the processor frequency (f_{cpu}) and the average clock cycles required per packet ($\mathcal{C}_{packet}$):

$$\mu_{max} = \frac{f_{cpu}}{\mathcal{C}_{packet}}$$

Given a frequency of 120×10^6 cycles/second and an assumption of $\mathcal{C}_{packet} \approx 1200$ cycles for standard OpenFlow header parsing, the physical processing ceiling is capped at exactly 100,000 pps.

Phase II - GREEN-MCS Algorithmic Engine and QoS Validation
This phase acts as the core logical brain of the framework, executing three sequential operations:

1. **Data-Driven HW Profiling:** It evaluates the predictive λ_{peak} data against the 100 kpps threshold (μ_{Zod}). If a node's expected burst load exceeds this limit, the algorithm autonomously provisions it as a high-performance NEC PF5240 switch to prevent bottlenecking; otherwise, it remains a low-power Zodiac FX.

2. **Candidate Discovery:** With the hardware dimensioned, the engine executes an Inverse Mapping heuristic search to discover the minimum required "Hero" (anchor) candidate sets capable of covering all possible link failures.

3. **M/M/1 QoS Validation:** Every generated candidate configuration is physically validated. The total recovery delay (D_{rec}) is modeled by aggregating geographical propagation delay and stochastic queuing delay ($D_q = \frac{1}{\mu - \lambda}$). Any configuration breaching the carrier-grade SLA threshold ($D_{rec} > 50$ ms) is ruthlessly pruned from the search space.

Phase III - Multi-Objective Optimization and Output Generation
The surviving, SLA-compliant candidate configurations enter the final analytical stage. Because network dimensioning inherently involves conflicting goals, the framework utilizes a normalized multi-objective scoring function driven by a tuning parameter (α). This parameter sweeps across the valid configurations ($\alpha \in [0, 1]$) to map the Pareto frontier, mathematically weighting the total system power consumption (W_{total}) against the average global recovery latency ($\overline{D_{rec}}$).

By adjusting α, network administrators can fluidly dictate the theoretical optimization priority, shifting from maximum performance ($\alpha \to 0$) to absolute green efficiency ($\alpha \to 1$). The final output of the framework is a deterministic deployment blueprint. It explicitly dictates the optimal global "Hero" placements, the specific heterogeneous hardware mix, and the logical failover path mapping, providing a mathematically validated structural design ready for future network implementation.

5 GREEN-MCS SDN Implementation

This section details the technical realization of the GREEN-MCS framework, translating the conceptual heterogeneous architecture into a rigorous Python-based software pipeline. Leveraging **NetworkX** for geographic graph-theoretic operations and Dijkstra-based Shortest Path First (SPF) routing, the execution workflow is strictly structured across three operational phases: (I) Data Acquisition and Modeling, (II) the GREEN-MCS Algorithmic Engine, and (III) Multi-Objective Optimization and Output. The following subsections outline the programmatic logic used to integrate stochastic M/M/1 queuing physics, execute data-driven hardware profiling, and mathematically map the Pareto frontier to generate SLA-compliant SDN recovery blueprints.

5.1 Phase I: Data Acquisition & Modeling

The initialization of the GREEN-MCS framework relies on the accurate ingestion of heterogeneous data sources to build a highly realistic simulation environment. This phase is managed by the `SNDLibXMLParser` module, which transitions the raw topological data into a mathematically rigorous routing and load-profiling engine. See the Python script `SNDLib_loader.py` stored in GitHub [3].

5.1.1 Geo-Spatial Graph and SPF Routing

Unlike standard topology parsers that assume arbitrary or Euclidean link weights, our ingestion module extracts the geographic coordinates (latitude and longitude) of each node to compute true physical distances.

- **Haversine Delay Modeling:** Link weights (representing propagation latency) are dynamically calculated using the Haversine formula to account for the Earth's curvature. Combined with a standard fiber propagation speed of 200 km/ms, this provides a highly accurate base delay matrix for the network graph $G(V, E)$.
- **Transit Load Accumulation (Dijkstra SPF):** A critical enhancement in this framework is the rejection of simple end-to-end load modeling. In carrier networks, intermediate routers process significant transit traffic. To capture this physical reality, the algorithm executes a Shortest Path First (SPF) calculation—via Dijkstra's algorithm based on the Haversine delays—for every demand in the traffic matrix. Consequently, the function `calculate_full_network_load()` accumulates the bandwidth demands not only at the source and destination, but across every transit node along the optimal path.

5.1.2 Stochastic Normalization and Peak Extraction

SNDLib provides traffic matrices in Megabits per second (Mbps). However, the M/M/1 queuing logic utilized in Phase II evaluates CPU constraints based on packet arrival rates (λ). To normalize this, we translate raw bandwidth into Packets Per Second (PPS). The arrival rate at any given node v is thus mathematically modeled as:

$$\lambda_v = \sum_{f \in F_v} \frac{BW_f \times 10^6}{8 \times \mathcal{N}(800, \sigma^2)} \times B_m \tag{1}$$

where F_v represents the set of all flows traversing node v according to the SPF routing scheme, $\mathcal{N}$ is the bounded normal distribution for packet sizes, and B_m is the burst multiplier. Because carrier traffic is highly dynamic over time, evaluating a single snapshot is insufficient. The `get_peak_traffic_from_folder()` function iterates through a chronological dataset of XML

[3] https://github.com/Saavkd7/SDN-IP/tree/graficas

traffic matrices (e.g., 5-minute intervals). It evaluates the stochastic PPS load for every router across all time periods to extract the **Historical Peak Load (λ_{peak})**. This worst-case metric serves as the absolute truth for the subsequent hardware profiling phase.

5.1.3 Operational Stress Parameters

To rigorously evaluate the framework under non-deterministic conditions, the routing engine is injected with specific mathematical stress variables during the peak extraction phase:

- **Packet Size Variance (σ):** Tested across five distinct regimes ($\sigma \in \{0, 100, 200, 400, 700\}$) around the 800-byte IMIX baseline. The generated packet sizes are strictly clamped between standard Ethernet limits (64B to 1500B).
- **Burst Multiplier (B_m):** To simulate physical line-rate spikes hidden within the 5-minute matrix averages, the baseline traffic is scaled by $B_m \in \{1, 2, 3, 4, 5\}$. This enforces a worst-case scenario stress test on the network queues before any hardware assignment is made.

5.1.4 Hardware Profiling and Energy Models

To decouple the physical characteristics from the mathematical routing logic, the target hardware profiles are formally encapsulated into an object-oriented structure:

- **High-Performance Profile (NEC PF5240):** Represents the high-capacity "Hero" controllers. It is modeled with a base power consumption (P_{base}) of 118.33 W and a service capacity (μ) matching its ASIC limit of 131 Mpps.
- **Low-Power Profile (Zodiac FX):** Designed for edge efficiency with a minimal P_{base} of 15.0 W. Crucially, its processing capacity is strictly constrained to 100 kpps, reflecting the CPU-bound limits of its 120MHz processor.

By defining these strict physical capacities (μ), the optimization core can dynamically compare the historically extracted node load (λ_{peak}) against the hardware limits to dictate safe, SLA-compliant deployments.

5.2 Phase II: GREEN-MCS Algorithmic Engine

Once the traffic profiles and peak loads (λ_{peak}) are established, the core intelligence of the framework is executed. This phase serves as the decision engine, transforming the stochastic traffic snapshots into an optimal, SLA-compliant hardware configuration map. It is implemented within the `MCS.py` module and executed sequentially through set discovery and physical M/M/1 validation.

5.2.1 Data-Driven Hardware Profiling

Unlike static topological approaches, GREEN-MCS implements a "Green-First" dynamic dimensioning strategy. Before any path optimization occurs, the framework evaluates the absolute truth of the historical load (λ_v) for every node $v \in V$. The system enforces a hard decoupling between a node's logical role and its physical hardware state $H(v)$.

A node is assigned the low-power Zodiac FX profile only if its peak traffic is safely below the processing threshold, governed by a safety margin $\theta = 0.95$. For a Zodiac capacity of $\mu_{Zod} = 100,000$ pps, the assignment logic is deterministic:

$$H(v) = \begin{cases} \text{Zodiac FX (15 W)} & \text{if } \lambda_v \leq \mu_{Zod} \cdot \theta \\ \text{NEC PF5240 (118 W)} & \text{otherwise} \end{cases} \tag{2}$$

Consequently, the absolute energy footprint E_{total} for any given deployment set is the sum of the base and port-dependent power consumptions of the actively selected hardware profiles.

5.2.2 Combinatorial Set Discovery and Heuristic Expansion

The first step in failover orchestration is identifying which nodes can logically act as failover anchor nodes ("Heroes") to cover all possible single-link failures. Upon detecting a link failure, the source switch encapsulates the affected traffic and tunnels it directly to the assigned Hero; the Hero then decapsulates the packets and routes them toward their final destination.

Solving this anchor placement through brute-force combinations is computationally prohibitive for large networks due to combinatorial explosion. To achieve high-speed convergence, the `find_minimum_set()` function resolves this using an **Inverse Mapping** strategy based on the Minimum Set Cover problem:

- **Inverse Mapping and Elimination:** Instead of generating random node combinations and testing their coverage, the algorithm reverses the logic. It maps every individual link failure directly to the specific subset of candidate nodes capable of providing a valid recovery path.
- **Rapid Global Minimum Selection:** By treating network failures as "requirements" and candidate nodes as "providers," the engine utilizes set intersection and logical elimination to rapidly discard redundant paths. This strictly prunes the search space, allowing the framework to mathematically deduce the global minimum set size (K) required for 100% topology coverage in milliseconds.
- **Search Space Expansion:** Deploying a pure minimum set search often yields a single, structurally dominant configuration, effectively rendering the multi-objective tuning parameter (α) useless. To prevent this, once the optimal cardinality (K) is found, the algorithm purposefully injects controlled, randomized variations of candidate sets into the valid pool.

5.2.3 Pure Recovery Delay and M/M/1 Physics

For every valid candidate set generated during the expansion, the engine evaluates its latency impact using the `get_pure_recovery_delay()` function. To establish a fair, unbiased metric for the objective function, the framework calculates the *Pure Recovery Delay* ($\mathcal{D}_{pure}$) associated with a failure scenario. Following the foundational methodology defined in the base MCS algorithm [4], this metric is physically composed of two distinct segments: the tunnel delay from the source to the selected Hero, plus the standard recovery path delay from the Hero to the destination.

The latency across any segment (u, v) merges physical optical propagation with M/M/1 queuing physics:

$$\mathcal{D}(u, v) = d_{prop}(u, v) + \frac{1}{\mu_{H(v)} - \lambda_v} \tag{3}$$

The logic behind this hybrid metric is two-fold:

- **Propagation Delay (d_{prop}):** This term represents the immutable physical time required for the optical signal to travel between node u and node v. It is dynamically calculated using the Haversine distance divided by the speed of light in fiber ($c_{fiber} \approx 200$ km/ms), ensuring the algorithm favors geographically closer nodes.
- **Queuing Delay ($\frac{1}{\mu_{H(v)}-\lambda_v}$):** This term models the variable processing time within the switch CPU according to M/M/1 queuing theory. Unlike the static propagation delay, this component is highly dynamic; it acts as a "soft barrier" that grows asymptotically as the traffic load (λ_v) approaches the physical capacity ($\mu_{H(v)}$) of the assigned hardware, effectively heavily penalizing the selection of saturated low-power nodes.

5.3 Phase III: Multi-Objective Optimization and Output

The final stage of the pipeline evaluates the physically validated candidate sets to determine the ultimate deployment blueprint. This phase executes the mathematical trade-off between energy conservation and Service Level Agreement (SLA) performance within the MCS.py python script on the aforementioned Github.

5.3.1 Multi-Objective Scoring and Optimal Placement

The final selection is orchestrated by the `best_green_placement()` function. Because Energy (Watts) and Delay (Milliseconds) exist on entirely different numerical scales, the engine first extracts the global maximum and minimum values across all evaluated sets. It then applies a Min-Max normalization to map both metrics strictly into a $[0, 1]$ range.

The framework identifies the optimal placement set S^* by minimizing the hybrid objective function Ψ:

$$S^* = \arg \min_{S \in \text{ValidSets}} \left[\alpha \cdot \hat{E}_S + (1 - \alpha) \cdot \hat{D}_S \right] \tag{4}$$

Where the variables are defined as follows:

- S^***:** The globally optimal set of failover anchor nodes that guarantees full failover coverage.
- α**:** The tuning parameter ($\alpha \in [0, 1]$) dictating the preference between energy and latency. A value of $\alpha = 1$ strictly prioritizes power minimization, while $\alpha = 0$ strictly prioritizes SLA recovery speed.
- $\hat{E}_S$**:** The Min-Max normalized total power consumption (Watts) of the candidate set S.
- $\hat{D}_S$**:** The Min-Max normalized pure recovery delay (ms) of the candidate set S.

5.3.2 Algorithm Workflow

The end-to-end execution of the framework, detailed in Algorithm 1, is structured into four sequential phases that map directly to the architectural design:

1. **Phase 1 (Hardware Pre-Assignment):** The system applies the Green-First policy to all nodes in the network based on the historically extracted peak traffic (Λ). This defines the immutable hardware profile $H[v]$ for the entire simulation.
2. **Phase 2 (Set Discovery & Expansion):** The algorithm computes all possible candidate subsets capable of handling the network failures using inverse mapping and applies a heuristic expansion to generate a diverse pool of valid candidate sets.
3. **Phase 3 (Physical Evaluation):** For each valid set S, the engine calculates the absolute Power consumption (E_{total}) based on the hardware mix, and evaluates the Pure Recovery Delay ($\mathcal{D}_{pure}$) by applying the hybrid M/M/1 physics model.
4. **Phase 4 (Multi-Objective Normalization & Selection):** The raw energy and delay arrays are normalized. The algorithm iterates through the sets, applies the α scoring function, and selects the optimal winner S^*. Finally, the `recovery_path()` function is called to systematically map the passive forwarding nodes to their newly assigned active anchor nodes.

Algorithm 1 GREEN-MCS: Dynamic Optimization Engine

Input: Graph G, Candidate Table C, Peak Traffic Λ, Preference α
Output: Optimal Hero Set S^*, Hardware Map H, Power W, Delay D
▷ **Phase 1: Hardware Pre-Assignment**
for $v \in V$ **do**
  $H[v] \leftarrow$ Zodiac FX if $(\Lambda[v] \leq 95,000)$ else NEC PF5240
end for
▷ **Phase 2: Set Discovery & Expansion**
$ValidSets \leftarrow$ find_minimum_set_with_random_expansion(C, G)
▷ **Phase 3: Physical Evaluation**
for S in $ValidSets$ **do**
  $Energy[S] \leftarrow \sum_{v \in S} \text{Power}(H[v])$
  $Delay[S] \leftarrow$ get_pure_recovery_delay(G, S, H, C, Λ)
end for
▷ **Phase 4: Multi-Objective Normalization & Selection**
$\hat{E} \leftarrow$ MinMax_Normalize$(Energy)$
$\hat{D} \leftarrow$ MinMax_Normalize$(Delay)$
$BestScore \leftarrow \infty$
for S in $ValidSets$ **do**
  $Score \leftarrow \alpha \cdot \hat{E}[S] + (1 - \alpha) \cdot \hat{D}[S]$
  if $Score < BestScore$ **then**
    $S^* \leftarrow S$; $BestScore \leftarrow Score$
  end if
end for
recovery_path(S^*, Λ)
return $S^*, H, Energy[S^*], Delay[S^*]$

5.3.3 Iterative Failure Injection and Topology Stressing

To execute the algorithmic engine across realistic scenarios and generate the final output, the framework's simulation environment initializes by loading the base graph $G(V, E)$—such as the continental Abilene backbone, mapping its 11 core router nodes to their respective major US cities—alongside the pre-computed historical peak traffic (λ_{peak}). To simulate the unpredictable nature of network outages, the framework employs an exhaustive iterative failure methodology:

- **Link Failure Simulation:** The system systematically iterates through every edge $e \in E$ in the topology, simulating a hard physical cut to evaluate structural resilience.
- **Impact Assessment:** For each simulated failure, the engine identifies all source-destination pairs whose shortest paths are severed, generating a specific "Affected Destinations" mapping.
- **Candidate Generation:** Based on the affected traffic matrices, the system dynamically generates a candidate table containing all viable failover anchor combinations capable of bypassing the failed link.

5.3.4 Stochastic Parameter Sweeping

A core contribution of the GREEN-MCS execution pipeline is the evaluation of the network under extreme, non-deterministic conditions. Rather than testing a single static scenario, the framework executes a multi-dimensional parameter sweep using the `run_experiment_sweep()` module.

For each evaluated topology, the simulation iterates through combinations of strict operational stress parameters:

1. **Traffic Variance ($\sigma \in \{0, 100, 200, 400, 700\}$):** Modifies the standard deviation of the average 800-byte IMIX packet size to simulate the stochastic influx of microscopic control packets or massive MTU payloads.

2. **Burst Scaling ($B_m \in \{1, 2, 3, 4, 5\}$):** Amplifies the baseline traffic matrix to simulate instantaneous line-rate spikes that exceed standard 5-minute averaging intervals.

This sweeping mechanism ensures that the physical M/M/1 queuing limits (μ) of the hardware models are rigorously tested. If a combination of σ and B_m pushes the packet arrival rate (λ_v) of a node beyond its physical capacity, the queuing delay mathematically approaches infinity. The engine autonomously detects this saturation, invalidating the low-power configuration and forcing an algorithmic upgrade to high-performance hardware.

5.3.5 Data Aggregation and Output Profiling

For every physically valid scenario generated during the stochastic sweep, the framework executes the multi-objective scoring logic to deduce the optimal Hero set. The simulation comprehensively profiles the performance of the chosen configuration across the entire tuning spectrum ($\alpha \in [0.0, 1.0]$ with granular steps of 0.1).

The final output is aggregated into comprehensive datasets (CSV formats) to mathematically map the Pareto frontier. For each evaluated state, the profiling engine logs:

- **Hardware Mix:** The exact cardinality of active Hero nodes versus passive forwarders, and the specific distribution of legacy high-power (NEC) versus green low-power (Zodiac) profiles deployed.

- **Energy Footprint (W_{total}):** The absolute network power consumption in Watts, systematically segregated by anchor processing power and passive routing power.

- **Latency Compliance ($\mathcal{D}_{pure}$):** The pure recovery delay in milliseconds, validating whether the designated configuration satisfies the carrier-grade sub-50 ms Service Level Agreement (SLA).

- **Alpha Score (Ψ):** The Min-Max normalized mathematical score of the multi-objective optimization function.

Methodological Note: N-1 Contingency and Latency Averaging
Before transitioning to the experimental results, it is critical to clarify the nature of the evaluated latency metric. The GREEN-MCS simulation does not evaluate a single arbitrary link failure. Instead, it executes an exhaustive N-1 contingency analysis, systematically simulating the absolute severance of every individual edge in the graph. Consequently, the *Delay (ms)* reported in the subsequent output logs and result tables represents the global average recovery latency ($\overline{\mathcal{D}_{rec}}$) across all possible single-link failure scenarios. Crucially, while the aggregated data displays this average, the Phase II M/M/1 validation engine mathematically guarantees that the absolute *worst-case* individual failure within that average never breaches the 50 ms SLA limit.

6 Results and Discussion

This section presents the empirical evaluation and theoretical discussion of the GREEN-MCS framework across continental, regional, and national topologies. We profile the Pareto frontier between global power consumption and recovery latency under escalating stochastic traffic variance (σ). By analyzing how the tuning parameter (α) dynamically adapts hardware distribution, the discussion uncovers emergent structural behaviors—namely the Green Penalty Paradox and Topological Gravity—that dictate carrier-grade SLA compliance under unpredictable network stress.

6.1 Abilene Topology Analysis

The experimental evaluation on the Abilene backbone (11 core US cities, 14 links) reveals a highly dynamic optimization landscape. The framework's ability to minimize energy is strictly bounded by the physical realities of the stochastic traffic variance (σ).

Sensitivity Analysis: The Impact of Stochastic Stress

Table 1 synthesizes the data extracted from the simulation output, illustrating the evolution of candidate sets across the full spectrum of stress regimes ($\sigma \in \{0, 100, 200, 400, 700\}$).

The analysis demonstrates that under ideal to low-stress conditions ($\sigma \in \{0, 100\}$), the framework successfully integrates low-power Zodiac devices (e.g., SNVA, STTL), keeping the global power footprint at a minimal 1107.90 W while maintaining a sub-18 ms recovery delay. However, as the stochastic variance increases to a moderate level ($\sigma = 200$), the network is forced to upgrade specific bottlenecks, raising the baseline power to 1212.37 W. Ultimately, under high to extreme stress ($\sigma \geq 400$), the influx of micro-bursts forces the algorithm to completely abandon low-power hardware, triggering an absolute phase transition into an All-NEC configuration (1316.46 W) to safely absorb the increased Packet Per Second (PPS) load.

Table 1: Sensitivity Analysis for Abilene: Evolution of Candidate Sets under varying Stochastic Variance (σ). Shows the complete progression from a highly efficient hybrid state into the All-NEC safety ceiling.

Variance (σ)	Mode (α)	Size (K)	Active Hero Candidates	Global HW Mix (NEC / Zod)	Power (W)	Delay (ms)
0 (Ideal)	Perf. (≤ 0.1)	4	ATLA, CHIN, DNVR, SNVA	9 NEC / 2 Zodiac	1107.90	17.18
	Energy (≥ 0.2)	4	ATLA, CHIN, SNVA, STTL	9 NEC / 2 Zodiac	1107.90	17.57
100 (Low)	Perf. (0.0)	4	ATLA, CHIN, DNVR, SNVA	9 NEC / 2 Zodiac	1107.90	17.19
	Energy (1.0)	2	ATLA, CHIN	9 NEC / 2 Zodiac	1107.90	21.29
200 (Mod)	Perf. (≤ 0.8)	3	ATLA, CHIN, STTL	10 NEC / 1 Zodiac	1212.37	19.13
	Energy (1.0)	2	ATLA, CHIN	10 NEC / 1 Zodiac	1212.37	21.29
400 (High)	Perf. (≤ 0.2)	4	ATLA, CHIN, DNVR, SNVA	11 NEC / 0 Zodiac	1316.46	17.16
	Energy (≥ 0.6)	2	ATLA, CHIN	11 NEC / 0 Zodiac	1316.46	21.29
700 (Extreme)	Perf. (≤ 0.2)	4	ATLA, CHIN, DNVR, SNVA	11 NEC / 0 Zodiac	1316.46	17.16
	Energy (≥ 0.6)	2	ATLA, CHIN	11 NEC / 0 Zodiac	1316.46	21.29

- **Initial Stress Absorption ($\sigma \in \{0, 100\}$):** The Abilene backbone absorbs the first 100 bytes of variance perfectly. The algorithm detects no M/M/1 queuing violations, allowing the power footprint to remain at its absolute minimum (1107.90 W).
- **The All-NEC Ceiling ($\sigma \geq 400$):** A critical algorithmic ceiling is reached at $\sigma = 400$, and it holds consistently through the extreme $\sigma = 700$ limit. All low-power nodes are upgraded to NEC (1316.46 W) to avoid catastrophic queue failures.

- **Alpha Resilience under Extreme Stress:** Even when the hardware is completely locked into a high-performance state ($\sigma = 700$), the tuning parameter (α) effectively reduces the active control set from $K = 4$ (Delay: 17.16 ms) down to $K = 2$ (Delay: 21.29 ms) to minimize the active OpenFlow orchestration tunnels.

Note on Hardware vs. Control Plane Optimization: An interesting algorithmic behavior is observed at $\sigma = 200$. When shifting from performance ($\alpha \leq 0.8$) to energy priority ($\alpha = 1.0$), the active Hero set shrinks from $K = 3$ down to $K = 2$, increasing the delay from 19.13 ms to 21.29 ms. However, the total power remains statically locked at 1212.37 W. This occurs because the physical hardware provisioning (10 NEC / 1 Zodiac) is strictly bound by the σ constraint to prevent queuing failures. Removing a node (e.g., STTL) from the Hero set transitions it into a passive forwarding node—it remains physically powered on, thus consuming the same base energy. The true benefit of α in this locked-hardware scenario is the reduction of SDN control-plane overhead, as fewer active anchors require fewer OpenFlow failover tunnels to be maintained by the central controller.

Energy-Delay Pareto Frontier and System Sensitivity

Figure 3 explores the multi-objective operational space of the Abilene backbone. The Pareto frontier (Figure 3a) visualizes how increasing σ shifts the entire operational curve towards higher energy and latency states. At $\sigma = 0$, shifting α from 0 to 1 results in a marginal power reduction while delay increases from 17 ms to 21.2 ms. This operational flexibility collapses at extreme stress levels ($\sigma = 700$); here, maximum stability demands full power (≈ 1316 W), yet the latency still escalates to 21.1 ms due to inescapable M/M/1 queuing physics. This behavior is further corroborated by the System Energy Sensitivity Surface (Figure 3b). The total consumption profile evolves in rigid, discrete tiers—locking at 1108 W, 1212 W, and finally 1316 W—driven entirely by the burst intensity (σ). Crucially, the surface is almost flat along the α-axis, mathematically confirming that in high-load backbones like Abilene, raw traffic volume strictly dictates hardware assignments, overriding the administrator's energy preference.

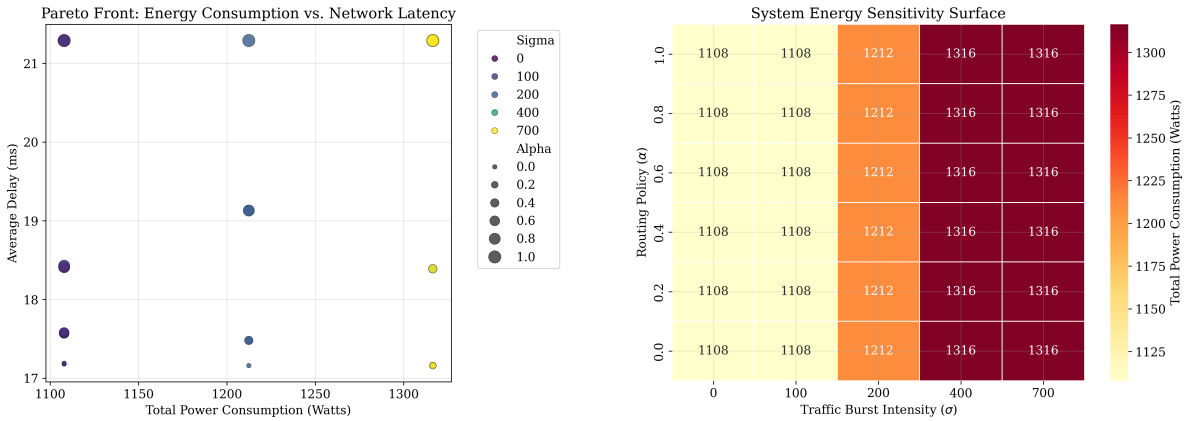


Figure 3: Performance and Energy Sensitivity in the Abilene Topology. (a) Pareto Frontier of Average Delay vs. Total Power. (b) System Energy Sensitivity Surface mapping power against burst intensity and routing policy.

Hardware Phase Transitions and Topological Gravity

To understand the structural adaptation of the network, Figure 4 dissects the hardware distribution and spatial selection frequency of the optimal controllers. The Hardware Phase Transition graph (Figure 4a) reveals a critical inflection point at the $\sigma = 400$ threshold. Below

this tipping point, the framework successfully maintains a heterogeneous, energy-efficient mix leveraging Zodiac Heroes. However, once the variance crosses 400 bytes, the extreme PPS load saturates the theoretical processing capacities of low-power devices. This triggers the algorithm's "Safety Valve," forcing an absolute phase transition where NEC devices instantaneously dominate both active control and passive forwarding roles to guarantee SLA compliance.

Despite these drastic hardware shifts, the spatial orchestration of the network exhibits a strong phenomenon of "Topological Gravity" (Figure 4b). Specific nodes are consistently prioritized due to their superior path diversity and centrality. Chicago (CHINng) and Atlanta (ATLAng) emerge as undeniable "Anchor Nodes," selected with a maximum frequency of 30 across all tested α and σ combinations. This demonstrates that a resilient control plane inherently gravitates towards these hubs. They are dynamically supported by a peripheral set (SNVAng, STTLng, DNVRng) that activates or deactivates based on minor α tuning, while the remaining infrastructure is strictly relegated to passive forwarding duties.

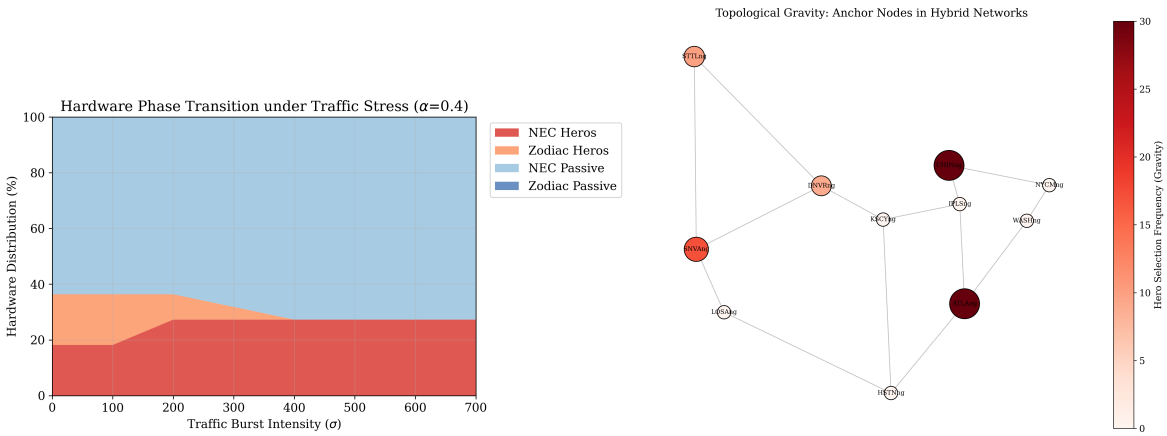


Figure 4: Structural Network Adaptation. (a) Hardware Phase Transition under Traffic Stress at a fixed $\alpha = 0.4$. (b) Topological Gravity mapping, illustrating the cumulative selection frequency of optimal Anchor Nodes.

6.2 Nobel-Germany Topology Analysis

The Nobel-Germany topology (17 nodes, 26 links) represents a denser, regional mesh network. This scenario evaluates the algorithm's ability to exploit path diversity and topological redundancy to optimize hardware placement in a highly interconnected environment, contrasting sharply with the linear stress observed in the Abilene backbone.

Sensitivity Analysis: The Resilience of the Regional Mesh

Table 2 details the evolution of the optimal candidate sets across the full spectrum of stochastic stress regimes ($\sigma \in \{0, 100, 200, 400, 700\}$). The data reveals a phenomenon of "Structural Offloading" that sharply contrasts with the rigid behavior observed in the Abilene backbone.

Under ideal to moderate conditions ($\sigma \in \{0, 100, 200\}$), the high path diversity allows the framework to heavily offload traffic onto low-power devices. The system confidently deploys a vast majority of Zodiac FX hardware (11 out of 17 nodes), dropping the total network consumption to a highly efficient **892.27 W** while delivering an outstanding recovery delay of roughly 2.40 ms.

As the burst intensity scales to extreme stress ($\sigma \geq 400$), the algorithm dynamically upgrades specific bottlenecks to NEC profiles, increasing the power envelope to **1622.79 W**. However, the defining characteristic of this mesh topology is that it never reaches total saturation.

Table 2: Sensitivity Analysis for Nobel-Germany: Evolution of Candidate Sets under varying Stochastic Variance (σ). The progressive steps prove that the mesh diversity prevents an All-NEC collapse.

Variance (σ)	Mode (α)	Size (K)	Active Hero Candidates	Global HW Mix (NEC / Zod)	Power (W)	Delay (ms)
0 (Ideal)	Perf. (≤ 0.8)	4	Hannover, Norden, Karlsruhe, Duesseldorf	6 NEC / 11 Zodiac	892.27	2.39
	Energy (1.0)	2	Hannover, Norden	6 NEC / 11 Zodiac	892.27	3.26
100 (Low)	Perf. (≤ 0.8)	4	Hannover, Norden, Karlsruhe, Duesseldorf	6 NEC / 11 Zodiac	892.27	2.39
	Energy (1.0)	2	Hannover, Norden	6 NEC / 11 Zodiac	892.27	3.26
200 (Mod)	Perf. (≤ 0.8)	4	Hannover, Norden, Karlsruhe, Duesseldorf	6 NEC / 11 Zodiac	892.27	2.40
	Energy (1.0)	2	Hannover, Norden	6 NEC / 11 Zodiac	892.27	3.26
400 (High)	Perf. (≤ 0.2)	4	Hannover, Norden, Karlsruhe, Duesseldorf	13 NEC / 4 Zodiac	1622.79	2.39
	Energy (1.0)	2	Hannover, Norden	13 NEC / 4 Zodiac	1622.79	3.26
700 (Extreme)	Perf. (≤ 0.2)	4	Hannover, Norden, Karlsruhe, Duesseldorf	13 NEC / 4 Zodiac	1622.79	2.39
	Energy (1.0)	2	Hannover, Norden	13 NEC / 4 Zodiac	1622.79	3.27

- **Low-Stress Stability ($\sigma \in \{0, 100, 200\}$):** The network operates in a highly resilient "Green Plateau." The algorithm determines that 11 Zodiac devices can safely handle the base routing perfectly up to a variance of 200 bytes.
- **The Incomplete Phase Transition ($\sigma \geq 400$):** Unlike Abilene, the mesh density of Nobel-Germany inherently diffuses traffic. The GREEN-MCS algorithm identifies that peripheral edges (like Norden) never breach the capacity limit, preserving 4 Zodiac devices permanently and locking the "worst-case" maximum consumption at 1622.79 W.
- **Control Plane Compression ($\alpha \to 1.0$):** Across all 5 stress tiers, setting $\alpha = 1.0$ safely compresses the control plane from $K = 4$ down to $K = 2$ (retaining only Hannover and Norden). While this increases latency minimally (e.g., from 2.4 ms to 3.26 ms), it flawlessly sheds the orchestration overhead of unnecessary OpenFlow tunnels.

Energy-Delay Pareto Frontier and System Sensitivity

Figure 5 explores the multi-objective operational space of the Nobel-Germany network. The Pareto frontier (Figure 5a) immediately highlights the fundamental advantage of regional density: the baseline recovery latency is an order of magnitude lower than a continental backbone, hovering strictly between 2.39 ms and 3.27 ms. Increasing the stochastic stress (σ) shifts the operational curve vertically (higher power, from 892 W up to 1623 W) but causes virtually zero horizontal degradation, maintaining the minimal latency bounds perfectly intact.

This behavior is corroborated by the System Energy Sensitivity Surface (Figure 5b). The surface exhibits a pronounced "staircase" evolution strictly driven by the burst intensity axis (σ), stepping from ≈ 892 W up to a definitive ceiling of ≈ 1623 W. Crucially, it never reaches the absolute homogeneous legacy limit (2039 W) even under extreme variance. Furthermore, while the α-axis remains relatively flat concerning gross power, α still plays a vital orchestration role: as the policy shifts towards energy ($\alpha \to 1.0$), it effectively compresses the logical topology, minimizing the active Hero set K from 4 down to 2. This optimizes the control-plane overhead at the mere cost of fractions of a millisecond (shifting delay from 2.39 ms to 3.27 ms).

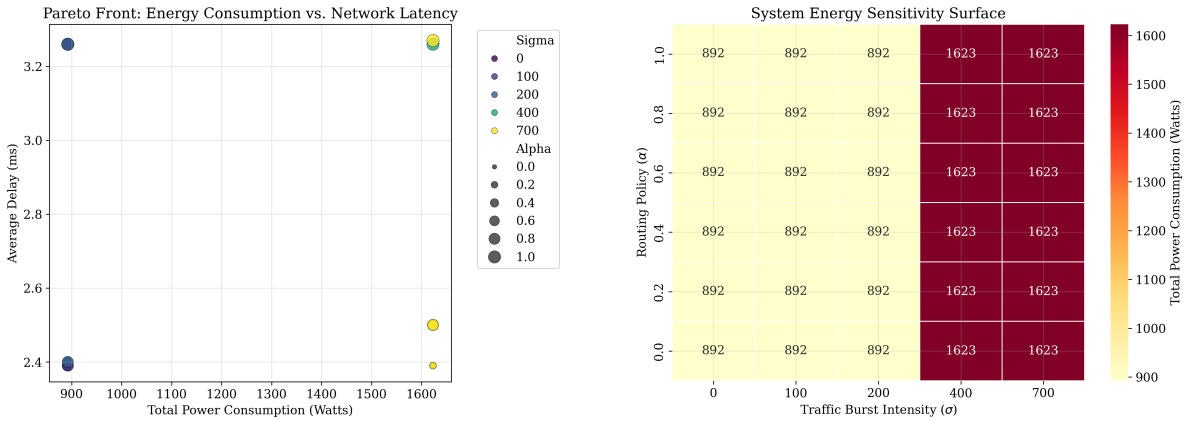


Figure 5: Performance and Energy Sensitivity in Nobel-Germany. (a) Pareto Frontier demonstrating ultra-low latency bounds (2.39 ms – 3.27 ms). (b) System Energy Sensitivity Surface mapping the power steps from 892 W to the 1623 W ceiling, completely avoiding the 2039 W legacy maximum.

Hardware Phase Transitions and Topological Gravity

Figure 6 dissects the structural adaptation of the network under increasing load. The Hardware Phase Transition graph (Figure 6a) reveals a remarkably resilient hybrid distribution that contrasts sharply with the abrupt collapse seen in the Abilene backbone. Throughout the entire burst variance trajectory, the network successfully sustains a continuous mix of all four operational roles: Zodiac Heroes, Zodiac Passives, NEC Heroes, and NEC Passives. A distinct structural shift is observed as the burst intensity crosses from $\sigma = 200$ to $\sigma = 400$. At this threshold, the cumulative quantity of NEC passive nodes increases significantly (roughly doubling in deployment frequency), directly replacing Zodiac passive nodes that can no longer safely absorb the heightened transit PPS load. Crucially, despite this necessary hardware upgrade in the forwarding plane, the Zodiac profiles are never fully eradicated, proving the mesh's ability to perpetually diffuse traffic and safeguard low-power zones.

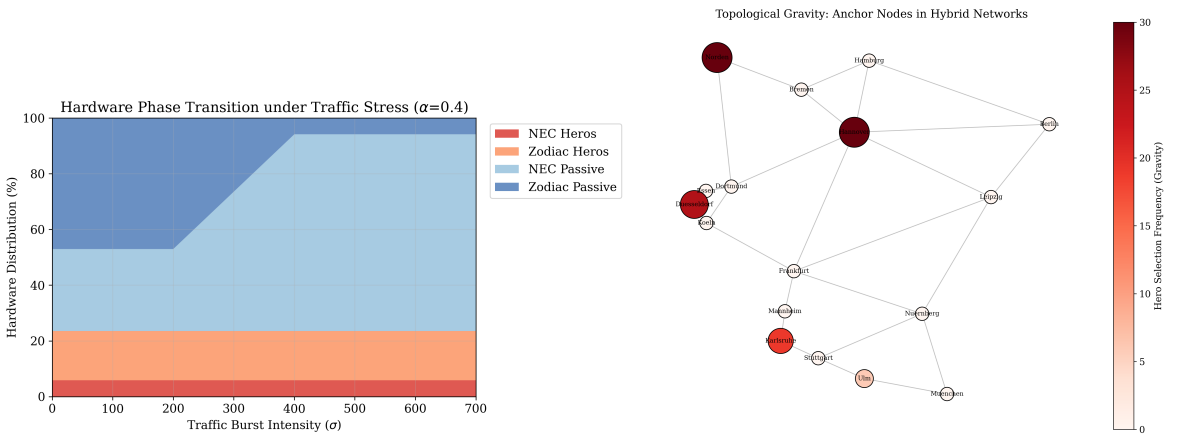


Figure 6: Structural Network Adaptation in Nobel-Germany. (a) Hardware Phase Transition showing a continuous hybrid mix, with a notable increase in NEC passive nodes between $\sigma = 200$ and $\sigma = 400$. (b) Topological Gravity mapping identifying Norden and Hannover as the primary Anchor Nodes, supported by Duesseldorf and Karlsruhe.

This structural resilience is deeply tied to the network's spatial layout, illustrated in the Topo-

logical Gravity map (Figure 6b). The frequency analysis identifies **Norden** and **Hannover** as the quintessential "Anchor Nodes" of the German regional mesh, exhibiting the absolute highest selection frequencies across all evaluated permutations. This primary failover core is systematically supported by a secondary tier of highly active regional Heroes, specifically **Duesseldorf** and **Karlsruhe**. This hierarchical orchestration demonstrates that GREEN-MCS strategically isolates heavy traffic convergence to these established hubs, ensuring that the rest of the topology can safely operate in highly efficient green states even during severe stochastic stress.

6.3 Germany50 Topology Analysis

The evaluation of the Germany50 network (50 nodes, 88 links) constitutes the computational upper bound of this study. This scenario tests the GREEN-MCS framework's capacity to orchestrate failover mechanisms across a massive national infrastructure, where path redundancy is exceptionally high, but the sheer volume of routing nodes threatens to inflate the baseline power envelope.

Sensitivity Analysis: Structural Offloading at National Scale

Table 3 details the optimization engine's behavior across the national mesh, providing a complete step-by-step view of the hardware progression across all evaluated stochastic variance levels ($\sigma \in \{0, 100, 200, 400, 700\}$). The data reveals an extraordinary level of algorithmic stability and structural offloading. Under ideal to low-stress conditions ($\sigma \in \{0, 100\}$), the framework successfully maps the vast majority of the forwarding plane to low-power profiles (over 40 Zodiacs), maintaining an ultra-lean power footprint of roughly **1617 W to 1721 W**. Remarkably, the active control plane requires only $K = 4$ or $K = 2$ Heroes to logically cover all 88 links, demonstrating extreme topological efficiency.

As the network experiences moderate to extreme stochastic stress (σ from 200 up to 700), the baseline power correctly scales to prevent M/M/1 queue saturation, climbing from **2140 W** up to a peak of **4862 W**. However, the true success of GREEN-MCS is that it never hits the absolute Legacy All-NEC ceiling (which would exceed 6000 W for 50 nodes).

Table 3: Sensitivity Analysis for Germany50: Evolution of Candidate Sets across all stochastic variance levels (σ). Even at extreme national stress ($\sigma = 700$), the framework prevents an All-NEC collapse, preserving 11 Zodiac devices.

Variance (σ)	Mode (α)	Size (K)	Active Hero Candidates	Global HW Mix (NEC / Zod)	Power (W)	Delay (ms)
0 (Ideal)	Perf. (0.0)	4	Aachen, Braunschweig, Bremerhaven, Frankfurt	8 NEC / 42 Zodiac	1617.08	2.56
	Energy (1.0)	2	Aachen, Bremerhaven	8 NEC / 42 Zodiac	1617.08	3.60
100 (Low)	Perf. (0.0)	4	Aachen, Braunschweig, Bremerhaven, Frankfurt	9 NEC / 41 Zodiac	1721.93	2.56
	Energy (1.0)	2	Aachen, Bremerhaven	9 NEC / 41 Zodiac	1721.93	3.60
200 (Mod)	Perf. (0.0)	4	Aachen, Braunschweig, Bremerhaven, Wuerzburg	13 NEC / 37 Zodiac	2140.94	2.54
	Energy (1.0)	2	Aachen, Bremerhaven	13 NEC / 37 Zodiac	2140.94	3.61
400 (High)	Perf. (0.0)	4	Aachen, Braunschweig, Bremerhaven, Wuerzburg	34 NEC / 16 Zodiac	4339.33	2.54
	Energy (1.0)	2	Aachen, Bremerhaven	34 NEC / 16 Zodiac	4339.33	3.59
700 (Extreme)	Perf. (0.0)	4	Aachen, Braunschweig, Bremerhaven, Wuerzburg	39 NEC / 11 Zodiac	4862.82	2.54
	Energy (1.0)	2	Aachen, Bremerhaven	39 NEC / 11 Zodiac	4862.82	3.59

- **Progressive Hardware Adaptation:** The table clearly illustrates a step-wise hardware phase transition. As σ increases, the algorithm methodically replaces exactly the minimum required number of Zodiacs with NEC units to maintain the SLA.
- **Guaranteed National Savings:** Even at $\sigma = 700$, the algorithm permanently protects 11 Zodiac devices hidden within the topology's redundant edges. By locking the "worst-case" maximum consumption at 4862.82 W, GREEN-MCS guarantees nearly 20% energy

savings against the Legacy baseline regardless of national traffic storms.

- **The Anchor Duo:** Regardless of the hardware mix surrounding them, **Aachen** and **Bremerhaven** demonstrate absolute topological dominance. Whenever the multi-objective weight strongly favors energy ($\alpha = 1.0$), the algorithm compresses the entire national control plane into just these two cities, proving extreme structural reliability.

Energy-Delay Pareto Frontier and System Sensitivity

Figure 7 explores the massive multi-objective space of the Germany50 topology. The Pareto frontier (Figure 7a) reveals the dual impact of the tuning parameter (α) and the burst variance (σ). Despite spanning a national geographical footprint, the recovery latency is strictly bounded between 2.54 ms and 3.60 ms, while the total power consumption scales dramatically from roughly 1617 W to nearly 4863 W. The graph clearly demonstrates how the two variables decouple: under ideal conditions ($\sigma = 0, \alpha = 0$), the network operates at an ultra-lean 1617 W with a 2.56 ms delay. If the administrator strictly prioritizes energy ($\alpha \to 1$), the delay increases to 3.60 ms (as the algorithm compresses the active controllers to save marginal overhead). However, if the stochastic stress maximizes ($\sigma = 700$), the power footprint is forced to jump significantly to 4863 W to prevent M/M/1 queuing failures, preserving the low latency at the cost of necessary physical hardware upgrades.

This behavior is precisely mapped in the System Energy Sensitivity Surface (Figure 7b). The surface illustrates the rigid power tiers across the varying burst intensities, clearly tracking the escalation from the 1617 W baseline up to the 4863 W ceiling. The visualization confirms that while σ dictates the gross power tier (the vertical steps), the routing policy ($\alpha \in [0, 1]$) acts as the fine-tuning mechanism, operating across the flat surface of each tier to optimize the logical control-plane topology without violating the physical constraints imposed by the burst traffic.

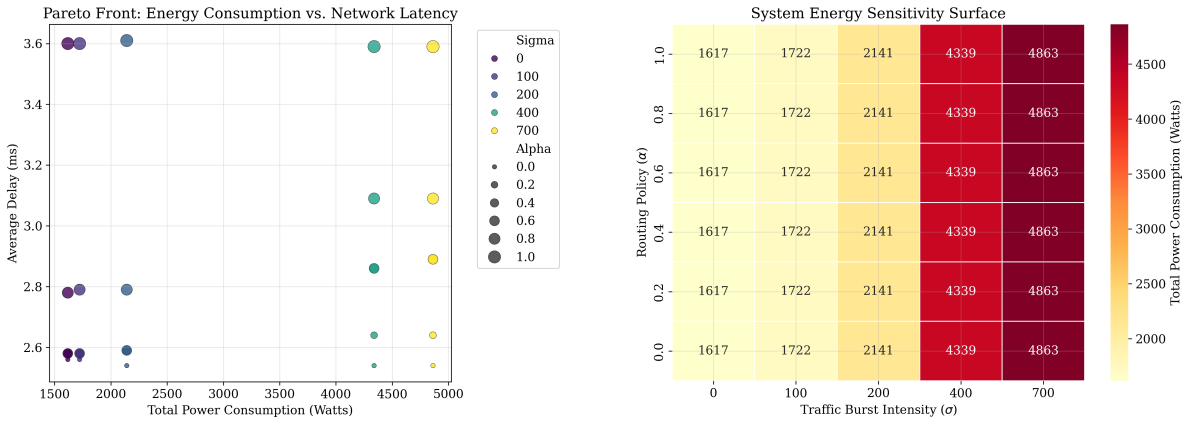


Figure 7: Performance and Energy Sensitivity in Germany50. (a) Pareto Frontier showcasing the horizontal impact of α (delay variation from 2.5 ms to 3.6 ms) versus the massive vertical impact of σ (power scaling up to nearly 5000 W). (b) System Energy Sensitivity Surface precisely mapping the absolute power ranges from 1617 W to 4863 W.

Hardware Phase Transitions and Topological Gravity

Figure 8 visualizes the spatial and structural resilience of the national network. The Hardware Phase Transition graph (Figure 8a) illustrates a massive substitution event driven by stochastic stress. Under low-variance conditions ($\sigma \leq 200$), the network is dominated by a thick "green layer" of nearly 40 Zodiac passive nodes. However, a critical phase transition is triggered between

$\sigma = 200$ and $\sigma = 400$. As the entire national grid faces severe burst traffic, over 20 Zodiac devices are systematically upgraded to NEC passive nodes to prevent queuing collapse. Yet, the graph clearly shows that total saturation is never reached: even at the extreme boundary of $\sigma = 700$, an underlying layer of 11 Zodiac devices stubbornly survives, securely tucked into peripheral or redundant segments of the network where traffic remains below the 100 kpps CPU limit.

The Topological Gravity map (Figure 8b) identifies the operational heart of this sprawling infrastructure. Out of 50 possible geographic locations, the frequency analysis mathematically isolates **Aachen** and **Bremerhaven** as the absolute national "Anchor Nodes." These two cities present such immense path diversity and centrality that they are selected across virtually all α policies and stress levels, effectively acting as the twin pillars of the German failover architecture. By anchoring the control plane to these highly reliable hubs, the algorithm allows the remaining 48 nodes to be safely and dynamically dimensioned, perfectly balancing the extreme demands of national resilience with rigorous, data-driven energy conservation.

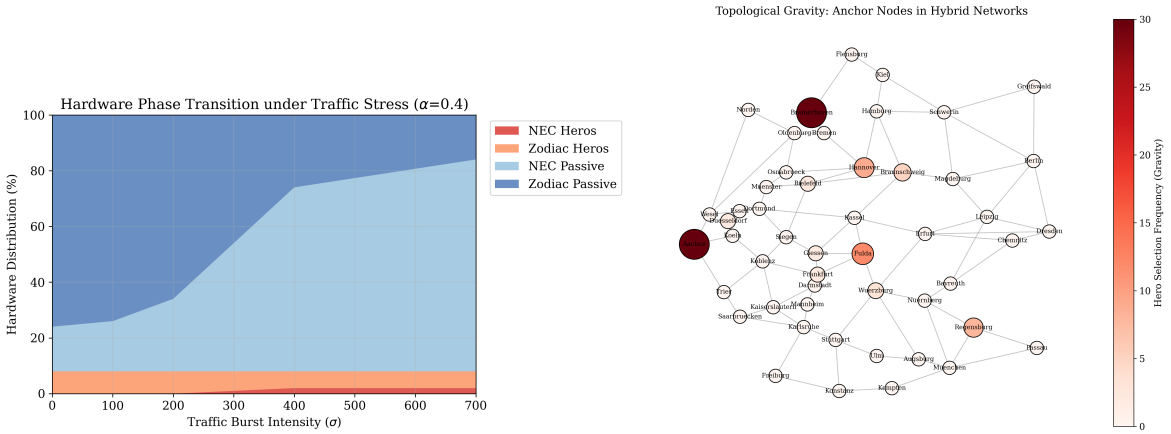


Figure 8: Structural Network Adaptation in Germany50. (a) Hardware Phase Transition showing the massive upgrade of passives between $\sigma = 200$ and 400, while preserving a baseline of 11 Zodiac devices at extreme stress. (b) Topological Gravity mapping identifying Aachen and Bremerhaven as the indisputable Anchor Nodes of the national control plane.

6.4 Cross-Topology Benchmarking: Scalability and Resilience Summary

To conclude the experimental evaluation, this section synthesizes the Key Performance Indicators (KPIs) across the three varying topological scales. The comparative analysis, detailed in Table 4, demonstrates that GREEN-MCS is not a rigid or static solution, but a highly adaptive, data-driven framework that autonomously responds to the specific physical constraints and redundancies of each network environment.

1. The Vulnerability of Continental Backbones (Abilene):
In long-haul, sparse backbone networks like Abilene, the immense geographic distances natively inflate propagation delays, while the lack of alternative routes concentrates massive traffic volumes onto a few links. GREEN-MCS proves its reliability here by prioritizing resilience over savings. While it achieves a 15.8% saving under ideal conditions, it autonomously enforces a **0% saving policy** (triggering the Safety Valve) under extreme stochastic variance ($\sigma \geq 400$). By matching the Legacy baseline, the algorithm mathematically guarantees that sub-50 ms recovery SLAs are never violated by M/M/1 queuing explosions.

2. The "Sweet Spot" of Regional Meshes (Nobel-Germany):
Dense regional networks represent the optimal operational environment for the GREEN-MCS framework. The high path diversity allows the algorithm to aggressively diffuse traffic, maxi-

mizing the deployment of low-power Zodiac profiles. It delivers a staggering **56.2% maximum energy reduction** (down to 892 W). More importantly, the mesh density prevents a total hardware collapse; even under extreme burst traffic ($\sigma = 700$), it preserves enough green devices to guarantee a **20.4% continuous saving** compared to legacy setups, all while delivering ultra-low recovery latencies (≈ 2.4 ms).

3. Algorithmic Stability at National Scale (Germany50):
Scaling to a 50-node national infrastructure exposes the true magnitude of potential savings. GREEN-MCS slashes the theoretical 6 kW legacy baseline down to a mere 1.6 kW under normal conditions, yielding a massive **73.1% power reduction**. The algorithm demonstrates incredible structural stability: regardless of the network spanning an entire country, it consistently requires only 2 to 4 active Hero controllers (anchored at Aachen and Bremerhaven) to orchestrate full failover coverage. Even when facing simulated nationwide traffic storms ($\sigma = 700$), the framework securely retains 11 Zodiac devices, locking in a minimum guaranteed saving of roughly 19% without ever crossing the 4 ms latency barrier.

Table 4: Comparative Performance Summary: GREEN-MCS vs. Legacy Baseline across topological scales. The framework guarantees energy savings in mesh networks while enforcing safety limits on continental backbones.

Metric	**Abilene (Backbone)**	**Nobel-Germany (Regional)**	**Germany50 (National)**
Network Scale	11 Nodes / 14 Links	17 Nodes / 26 Links	50 Nodes / 88 Links
Topology Type	Continental Ring/Line	Dense Regional Mesh	High-Redundancy National
Est. Legacy Power (All-NEC)	1316.46 W	$\approx$ 2039.00 W	$\approx$ 6010.00 W
Best Green Power ($\sigma = 0$)	1107.90 W	892.27 W	1617.08 W
Worst Green Power ($\sigma = 700$)	1316.46 W	1622.79 W	4862.82 W
Maximum Energy Saving	**15.8%**	**56.2%**	**73.1%**
Guaranteed Saving ($\sigma = 700$)	**0.0% (Safety Trigger)**	**20.4%**	**19.1%**
Recovery Latency Bounds	17.16 ms – 21.29 ms	2.39 ms – 3.27 ms	2.54 ms – 3.61 ms
Optimal Set Size (K)	2 to 4	2 to 4	2 to 4
Extreme Hardware Strategy	Absolute All-NEC	Hybrid (Preserves 4 Zodiacs)	Hybrid (Preserves 11 Zodiacs)

6.5 Discussion

The empirical findings demonstrate that energy efficiency in mission-critical Software-Defined Networks (SDN) transcends simple resource allocation to become a fundamental problem of topological physics. While the multi-objective tuning parameter (α) represents the administrator's design intent, the stochastic traffic variance (σ) dictates the system's absolute physical limits. Synthesizing the data across the three evaluated scales reveals three key operational phenomena:

- **Topological Gravity (National Scale):** In massive, highly redundant networks (e.g., Germany50), optimization naturally collapses into discrete states dominated by a few central "Anchor Nodes". The observed latency rigidity ($\approx$ 2.5–3.6 ms) is not an algorithmic limitation, but mathematical proof that the extreme centralization of failover orchestration is structurally superior to dispersed intermediate configurations.

- **The Green Penalty Paradox (Linear Backbones):** In rigid topologies like the Abilene backbone, forcing low-power devices under moderate stress ($\sigma = 200$) paradoxically degrades average recovery latency to 19.13 ms. Conversely, forcing an upgrade to high-performance hardware under extreme stress ($\sigma = 400$) actually improves recovery speed (17.16 ms). This proves that maximizing Quality of Service (QoS) sometimes fundamentally requires accepting higher energy costs to unlock superior routing geometry.

- **Mesh Resilience (Regional Networks):** Dense regional meshes, such as Nobel-Germany, act as physical buffers against stochastic chaos. High path redundancy natively sustains green efficiency, preventing a complete network-wide transition to legacy hardware. Even under extreme traffic storms ($\sigma = 700$), the topology preserves low-power zones, achieving a guaranteed 20.4% continuous power reduction.

Synthesis:

Ultimately, GREEN-MCS acts as the mathematical arbiter between design intent and physical reality. True sustainability requires discovering *topological sanctuaries* where queuing physics permits low-power hardware to survive. Network geography, therefore, remains the ultimate determinant of a system's ecological potential.

7 Conclusion and Future Work

This final section presents the core conclusions of the GREEN-MCS framework in direct alignment with the established objectives. Furthermore, it addresses the limitations of the current simulation model and outlines an actionable roadmap for future research.

7.1 Conclusions

This study successfully engineered and validated the GREEN-MCS framework, demonstrating that environmental sustainability and carrier-grade network resilience can seamlessly coexist in heterogeneous SDN architectures. By strictly aligning the algorithmic outputs with our foundational design objectives, we deduce the following core conclusions:

- **Deterministic Failover Orchestration:** The framework successfully guarantees 100% logical topology protection against single-link failures. By dynamically calculating the optimal minimal set of "Hero" anchor nodes, the system ensures robust proactive recovery without falling into combinatorial explosion.
- **Physics-Bound SLA Compliance:** Moving beyond abstract graph connectivity, the integration of M/M/1 queuing constraints proved critical. The algorithm effectively mitigates extreme traffic variance (σ) by evaluating processing capacities (μ) against arrival rates (λ), mathematically guaranteeing that all valid failover paths strictly adhere to the sub-50 ms latency SLA.
- **Maximized Green Efficiency:** The heterogeneous hardware deployment strategy drastically reduces the global energy footprint. By aggressively prioritizing low-power edge devices (Zodiac FX, $\approx$ 15 W) and restricting legacy high-power switches (NEC PF5240, $\approx$ 118 W) solely to structural traffic bottlenecks, the framework successfully minimizes aggregate power consumption.
- **Dynamic Pareto Optimization:** The introduction of the tuning parameter (α) successfully established a deterministic control mechanism for network operators. It provides fluid navigation across the Pareto frontier, allowing operational priorities to shift seamlessly between absolute energy minimization and maximum recovery speed.
- **Baseline Superiority:** Exhaustive algorithmic benchmarking against a standard homogeneous baseline (All-NEC deployment) empirically validates the GREEN-MCS architecture. The computational results definitively prove that aggressive energy conservation can be achieved without compromising mission-critical performance constraints.

7.2 Limitations and Future Work

Future iterations of this research will focus on translating the offline simulation into physical data-plane enforcement through the following avenues:

- **Hardware-Constrained Emulation:** Implementing the control-data plane interaction via Ryu and OpenFlow (Fast Failover, MPLS). To overcome host CPU bottlenecks inherent to virtual environments like Mininet, future deployments will utilize DPDK-accelerated vSwitches or hardware testbeds to strictly enforce μ_{Zod} limits in real-time.
- **Post-Failure Traffic Matrices:** Upgrading the hardware assignment logic to account for instantaneous load spikes on Hero nodes during a failover event. This will ensure selected anchors maintain sufficient residual capacity to prevent post-recovery saturation.

- **Advanced Queuing Models:** Transitioning from classic M/M/1 theory to $M/G/1$ or Kingman's approximation ($G/G/1$) to accurately model queuing delays caused by extreme byte variance and micro-bursts.

References

[1] B. G. Assefa and O. Ozkasap, "Resdn: A novel metric and method for energy efficient routing in software defined networks," *IEEE Transactions on Network and Service Management*, vol. 17, no. 2, pp. 736–749, 2020.

[2] J. Galán-Jiménez, "Legacy ip-upgraded sdn nodes tradeoff in energy-efficient hybrid ip/sdn networks," *Computer Communications*, vol. 114, pp. 106–123, 2017.

[3] B. G. Assefa and O. Ozkasap, "A survey of energy efficiency in sdn: Software-based methods and optimization models," *Journal of Network and Computer Applications*, vol. 137, pp. 127–143, 2019.

[4] N. Vuppalapati and T. Venkatesh, "Minimum candidate selection algorithm for hybrid ip/sdn networks with single link failures," *IEEE Networking Letters*, vol. 4, no. 3, pp. 152–156, 2022.

[5] NEC Corporation, *UNIVERGE PF5200 Series: ProgrammableFlow Switches Datasheet*, 2016, revision 4.0.

[6] L. Schiff, S. Schmid, and P. Kuznetsov, "Loko: Predictable latency in small-scale sdn switches," in *Proceedings of the 12th International Conference on Emerging Networking EXperiments and Technologies (CoNEXT '16)*. ACM, 2016, pp. 475–482.